

Active Directory Infrastructure Deployment

Enterprise Domain Build, Group Policy Hardening & Client Onboarding — cis.net

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1. Project Overview

1.1 Objective

This document records the end-to-end deployment of an Active Directory domain environment within a self-built home lab, performed to develop and demonstrate practical enterprise infrastructure and security-administration skills relevant to a SOC Analyst role. The build covers domain controller promotion, organizational unit design, user account provisioning, Group Policy security hardening, DHCP services, and client domain join with post-configuration verification.

Network Topology

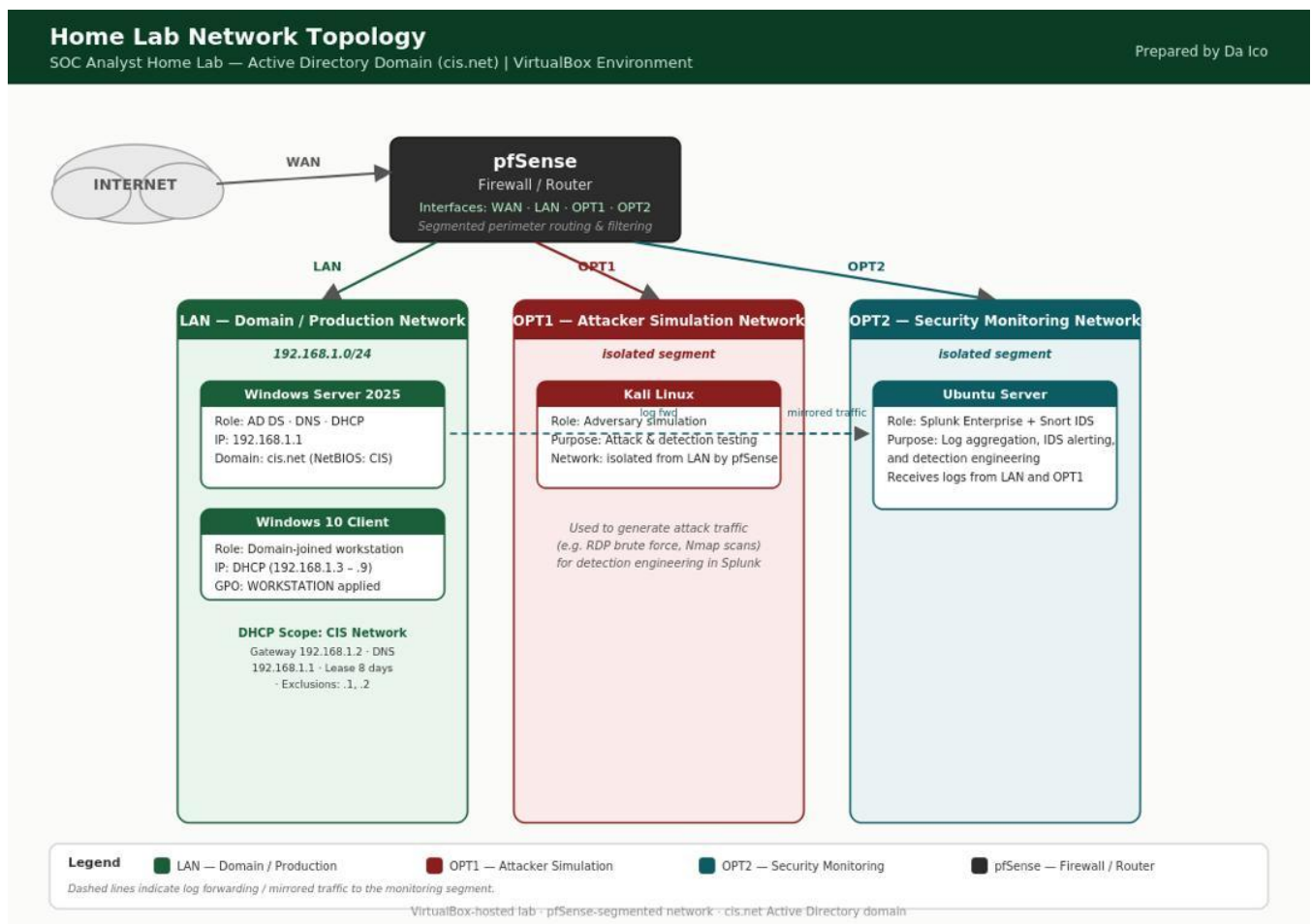


Figure 1: Network Topology

1.2 Lab Environment

Component	Role
pfSense	Firewall / router — segmented interfaces (WAN, LAN)
Windows Server 2025	Active Directory Domain Controller (AD DS, DNS, DHCP)
Windows 10	Domain-joined client workstation
Kali Linux	Attacker / adversary-simulation VM
Ubuntu Server	Splunk Enterprise + Snort IDS (log aggregation & detection)
VirtualBox	Virtualization platform hosting the full topology

2. Domain Controller Promotion & Organizational Unit Design

2.1 Promote Server to Domain Controller

1. Click the yellow flag in Server Manager at the top right → Promote this server to a domain controller
2. Select **Add a new forest**
3. Root domain name: **cis.net** → Next
4. Forest/Domain Functional Level: **Windows Server 2025**
5. Set a strong DSRM (Directory Services Restore Mode) password → Next
6. Acknowledge the DNS delegation warning → Next
7. Confirm NetBIOS name: **CIS** → Next
8. Leave default AD database, log file, and SYSVOL paths → Next
9. Review the configuration → Install (the server restarts automatically)

Post-reboot authentication uses **CIS\Administrator**.

2.2 Create Organizational Units (OUs)

1. Open **Active Directory Users and Computers** from Server Manager → Tools
2. **cis.net** → right-click → New → Organizational Unit
3. Create two OUs: one for computer objects, one for user objects
4. Naming convention: **Computer-OU** and **Users-org-OU**

Separating computer and user objects into dedicated OUs supports scoped Group Policy application and cleaner delegation of administrative control — a baseline practice for least-privilege AD design.

3. User Provisioning & Group Policy Security Hardening

3.1 Create User Accounts

1. On **cis.net** → right-click → New → User

First Name : Ambe
 Last Name : Peter
 Logon Name : ambe@cis.net
 Password : (-----) (strong password)
 Options : User should not change password

Object moved to: Users-org-OU

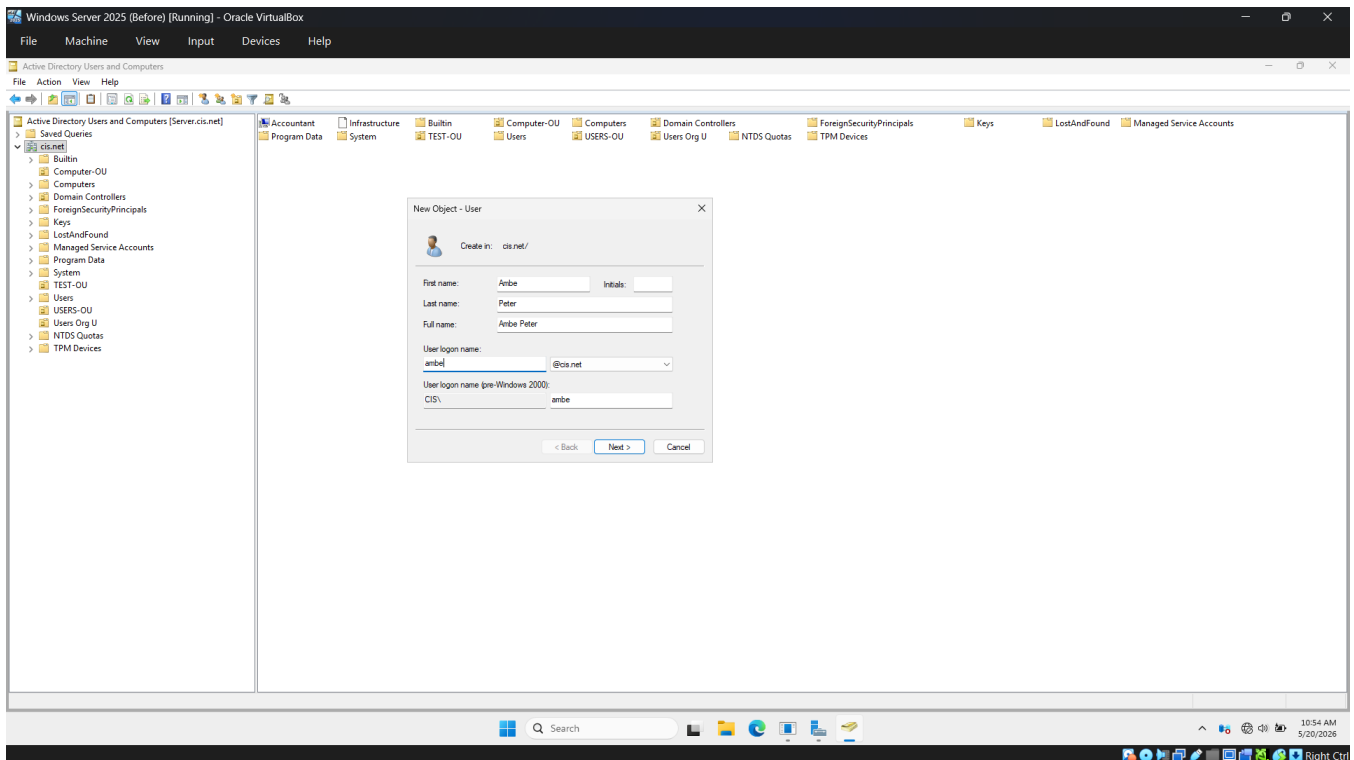


Figure 2: Creating Account

3.2 Configure Group Policy (GPO)

1. Server Manager → Manage → **Group Policy Management**
2. Expand **Forest: cis.net** → Domains → cis.net
3. Right-click **cis.net** → create and link a new GPO
4. Name: **WORKSTATION**
5. Right-click the GPO → **Edit**
6. Navigate to: **Computer Configuration > Policies > Windows Settings > Security Settings**

Policy Area	Configured Value
Account Lockout	Lock account after 5 failed logon attempts

Password Policy	Minimum 8 characters; complexity required
Audit Policy	Log success & failure for logon events
SOC Relevance Account lockout and audit-policy settings are what generate the raw telemetry a SOC depends on. Enabling success/failure auditing on logon events produces Windows Security Event IDs 4624/4625/4740 — the same event IDs used to build the RDP brute-force detection logic in this lab's Splunk instance. Configuring the baseline correctly at the domain level is the prerequisite for any downstream detection engineering work.	

4. DHCP Server Configuration

4.1 Install the DHCP Role

1. Server Manager → Manage → Add Roles and Features
2. Select DHCP Server → Add Features → Install
3. Click **Complete DHCP configuration** from the notification flag
4. Click **Commit** to authorize DHCP in Active Directory
5. Reboot

4.2 Create a DHCP Scope

1. Server Manager → Manage → DHCP → expand **server.cis.net**
2. Right-click IPv4 → New Scope
3. Name: **CIS Network**

Setting	Value
Start IP Address	192.168.1.1
End IP Address	192.168.1.9
Subnet Mask	255.255.255.0
Lease Duration	8 Days
Default Gateway	192.168.1.2
DNS Server	192.168.1.1
Exclusions	192.168.1.1, 192.168.1.2

Activate the scope (**Yes**) → Finish.

5. Windows 10 Client Domain Join

5.1 Configure DNS on the Client

On Windows 10 clients:

1. Go to **ncpa.cpl** → right-click the adapter → Properties
2. Open IPv4 → set: **Obtain DNS Server Address automatically**

Verify DNS Resolution

```
nslookup cis.net
ping cis.net
ipconfig /all
```

```
Windows 10 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

C:\Users\ambe>nslookup cis.net
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ambe>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\ambe>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-I17DHE4
Primary Dns Suffix . . . . . : cis.net
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search list. . . . . : cis.net

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : cis.net
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-26-EA-50
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::1620:f8f8:7970:8de7%3(Preferred)
IPv4 Address. . . . . : 192.168.1.4(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Wednesday, May 26, 2026 3:21:47 AM
Lease Expires . . . . . : Thursday, May 28, 2026 3:21:47 AM
Default Gateway . . . . . : 192.168.1.2
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 181187623
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-9F-A3-A4-08-00-27-26-EA-50
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\ambe>
```

Figure 3: Verifying Connectivity between ddomain controller and domain

5.2 Join the Domain

1. Right-click **This PC** → Properties → Rename this PC (advanced)
2. Click **Change** → select Domain
3. Enter: **cis.net** → OK
4. Provide domain-join credentials when prompted
5. Credentials: **ambe@cis.net** + password (-----)
6. Click **OK** on the welcome message → Restart

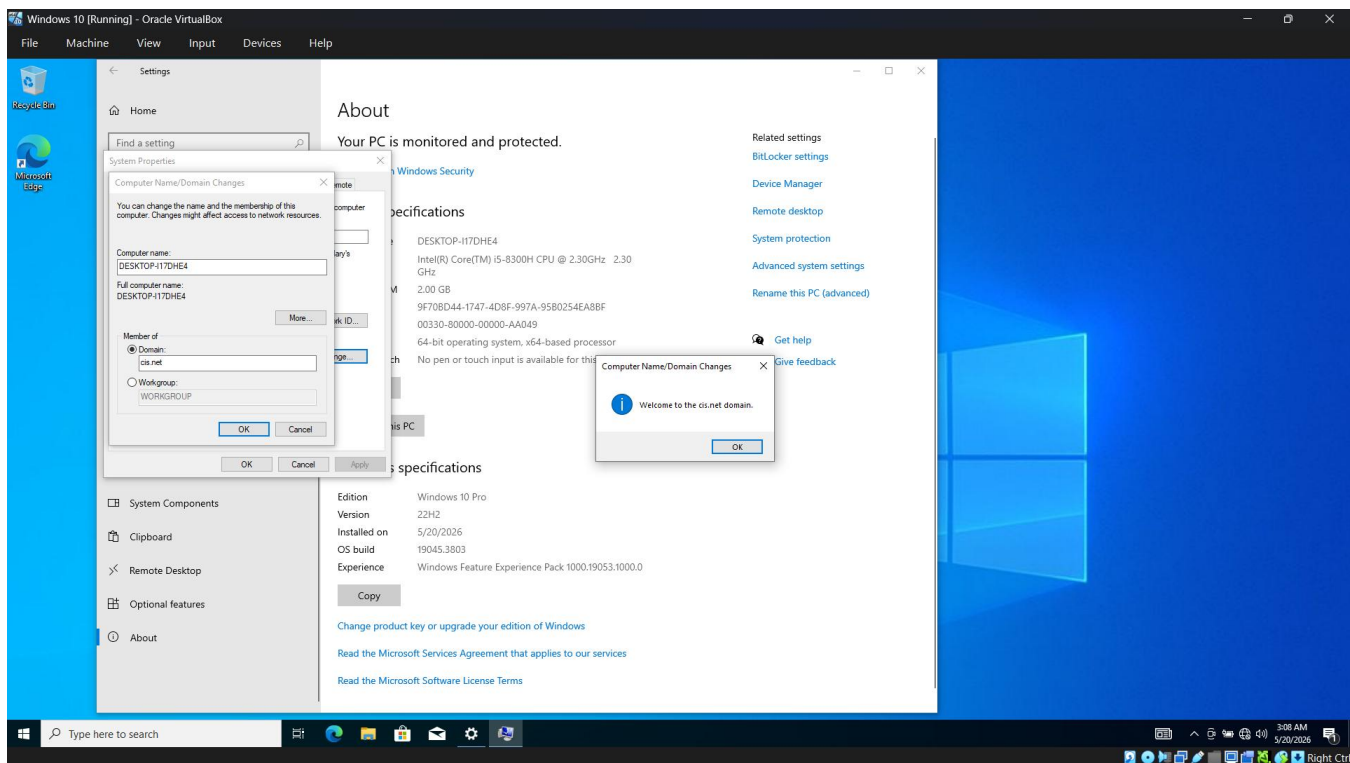


Figure 4: Successful connectivity to domain

6. Domain Login & Verification

6.1 Log In with a Domain Account

1. Click **Other user** on the login screen
2. Enter: **CIS\ambe**
3. Enter password → Enter

6.2 Verify Domain Membership

```
systeminfo | findstr /i cis.net
gpresult /r
```

Confirms the workstation reports domain membership and that the WORKSTATION GPO has applied successfully.

7. Quick Reference — Administrative Commands

Active Directory & DNS

dsa.msc	# Active Directory Users and Computers
dnsmgmt.msc	# DNS Manager
gpmc.msc	# Group Policy Management
dhcpgmt.msc	# DHCP Manager

```
dcdiag /v          # Diagnose DC health
netdom query fsmo   # Check FSMO role holders
repadmin /replsummary # Check AD replication
nslookup corp.local # Test DNS resolution
gpupdate /force     # Force Group Policy update
gpresult /r         # Show applied GPOs
```

8. Skills Demonstrated

1. Active Directory forest/domain deployment and DNS-integrated DC promotion
2. OU design supporting scoped GPO application and least-privilege delegation
3. Group Policy security baseline configuration (account lockout, password complexity, logon auditing)
4. DHCP role deployment, scope design, and AD authorization (DHCP Commit)
5. Windows client domain join, DNS troubleshooting, and post-join verification
6. Mapping of infrastructure configuration decisions to downstream security telemetry and detection engineering (Windows Event IDs 4624/4625/4740)

10. Lessons learned

- Kerberos authentication fails if the time difference between clients and Domain Controllers exceeds five minutes. The PDC Emulator must sync with a reliable external time source.
- Domain controllers, DNS servers, and DHCP servers require permanent, static IP addresses. Changing these IPs later disrupts the entire network topology.